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Display with light-emitting diodes

Abstract

A display may have an array of pixels each of which has a light-emitting diode such as an organic light-emitting diode. A drive transistor and an emission transistor may be coupled in series with the light-emitting diode of each pixel between a positive power supply and a ground power supply. The pixels may include first and second switching transistors. A data storage capacitor may be coupled between a gate and source of the drive transistor in each pixel. Signal lines may be provided in columns of pixels to route signals such as data signals, sensed drive currents from the drive transistors, and predetermined voltages between display driver circuitry and the pixels. The switching transistors, emission transistors, and drive transistors may include semiconducting-oxide transistors and silicon transistors and may be n-channel transistors or p-channel transistors.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6307322	12/2000	Dawson et al.	N/A	N/A
6466280	12/2001	Park et al.	N/A	N/A
6919871	12/2004	Kwon	N/A	N/A
7411571	12/2007	Huh	N/A	N/A
7737631	12/2009	Uhlig et al.	N/A	N/A
7978160	12/2010	Chung	N/A	N/A
8067775	12/2010	Miyairi et al.	N/A	N/A
8264426	12/2011	Chung	N/A	N/A
8289234	12/2011	Kim	345/204	G09G 3/3291
8421090	12/2012	Choi	N/A	N/A
8436342	12/2012	Park et al.	N/A	N/A
8455869	12/2012	Seo et al.	N/A	N/A
8530290	12/2012	Park et al.	N/A	N/A
8531618	12/2012	Koyama et al.	N/A	N/A
8581257	12/2012	Murai et al.	N/A	N/A
8711138	12/2013	Kawabe	345/212	G09G 3/3233
8890220	12/2013	Nathan et al.	N/A	N/A

8941697	12/2014	Nathan et al.	N/A	N/A
8993126	12/2014	Nowatari et al.	N/A	N/A
9129927	12/2014	Gupta et al.	N/A	N/A
9147719	12/2014	Kim et al.	N/A	N/A
9601557	12/2016	Yang et al.	N/A	N/A
9818344	12/2016	Lin et al.	N/A	N/A
10096622	12/2017	Gupta et al.	N/A	N/A
11462163	12/2021	Lin et al.	N/A	N/A
2001/0015778	12/2000	Murade et al.	N/A	N/A
2001/0043092	12/2000	McDaniel	N/A	N/A
2002/0180369	12/2001	Koyama	N/A	N/A
2005/0007316	12/2004	Akimoto et al.	N/A	N/A
2005/0206590	12/2004	Sasaki et al.	N/A	N/A
2005/0253531	12/2004	Matsumoto et al.	N/A	N/A
2006/0124944	12/2005	Kwak	N/A	N/A
2006/0208671	12/2005	Ozaki et al.	N/A	N/A
2007/0001205	12/2006	Kimura	N/A	N/A
2007/0040770	12/2006	Kim	N/A	N/A
2007/0236424	12/2006	Kimura	N/A	N/A
2009/0002590	12/2008	Kimura	N/A	N/A
2009/0002731	12/2008	Miyamoto et al.	N/A	N/A
2009/0114918	12/2008	Wang et al.	N/A	N/A
2009/0160741	12/2008	Inoue et al.	N/A	N/A
2009/0213046	12/2008	Nam	N/A	N/A
2009/0244057	12/2008	Seto	N/A	N/A
2009/0251496	12/2008	Saito et al.	N/A	N/A
2010/0019996	12/2009	Chun-Gi et al.	N/A	N/A
2010/0102313	12/2009	Miyairi et al.	N/A	N/A
2010/0134735	12/2009	Nakamura et al.	N/A	N/A
2010/0182223	12/2009	Choi	N/A	N/A
2010/0220117	12/2009	Kimura	N/A	N/A
2011/0031498	12/2010	Kimura et al.	N/A	N/A
2011/0199351	12/2010	Kurokawa	N/A	N/A
2011/0279430	12/2010	Koh	N/A	N/A
2011/0304311	12/2010	Takahashi et al.	N/A	N/A
2012/0001896	12/2011	Han et al.	N/A	N/A
2012/0006253	12/2011	Demma	N/A	N/A
2012/0026147	12/2011	Komiya	N/A	N/A
2012/0061668	12/2011	Miyairi et al.	N/A	N/A
2012/0147060	12/2011	Jeong	N/A	N/A
2012/0228596	12/2011	Fujita et al.	N/A	N/A
2012/0299978	12/2011	Chaji	N/A	N/A
2012/0305910	12/2011	Hsieh et al.	N/A	N/A
2013/0001564	12/2012	Choi et al.	N/A	N/A
2013/0002616	12/2012	Kim et al.	N/A	N/A
2013/0020795	12/2012	Giraldi	N/A	N/A
2013/0021549	12/2012	Fujita	N/A	N/A
2013/0109117	12/2012	Lee et al.	N/A	N/A
2013/0146866	12/2012	Kitagawa	N/A	N/A
2013/0162617	12/2012	Yoon et al.	N/A	N/A

2013/0168666	12/2012	Yan et al.	N/A	N/A
2013/0207957	12/2012	Lin et al.	N/A	N/A
2013/0214279	12/2012	Nishimura et al.	N/A	N/A
2013/0264570	12/2012	Cho et al.	N/A	N/A
2014/0018466	12/2013	Carlson et al.	N/A	N/A
2014/0049522	12/2013	Mathew et al.	N/A	N/A
2014/0138637	12/2013	Yang et al.	N/A	N/A
2014/0138651	12/2013	Oh	N/A	N/A
2014/0139510	12/2013	Han et al.	N/A	N/A
2014/0152685	12/2013	Iwaki	N/A	N/A
2014/0183491	12/2013	Shin et al.	N/A	N/A
2014/0197382	12/2013	Kim et al.	N/A	N/A
2014/0217373	12/2013	Youn et al.	N/A	N/A
2014/0217397	12/2013	Kwak et al.	N/A	N/A
2014/0231761	12/2013	Kim et al.	N/A	N/A
2014/0232956	12/2013	Kwon et al.	N/A	N/A
2014/0299884	12/2013	Park et al.	N/A	N/A
2014/0320544	12/2013	Kim	N/A	N/A
2014/0346475	12/2013	Cho et al.	N/A	N/A
2015/0001483	12/2014	Namkung	N/A	N/A
2015/0007741	12/2014	Nath et al.	N/A	N/A
2015/0049126	12/2014	Jung	N/A	N/A
2015/0054799	12/2014	Chang et al.	N/A	N/A
2015/0055047	12/2014	Chang et al.	N/A	N/A
2015/0055051	12/2014	Osawa et al.	N/A	N/A
2015/0109279	12/2014	Gupta et al.	N/A	N/A
2015/0123557	12/2014	Lee et al.	N/A	N/A
2015/0213757	12/2014	Takahama et al.	N/A	N/A
2015/0243220	12/2014	Kim et al.	N/A	N/A
2015/0243686	12/2014	Lee et al.	N/A	N/A
2015/0243718	12/2014	Kwon et al.	N/A	N/A
2015/0243720	12/2014	Kwon et al.	N/A	N/A
2015/0248856	12/2014	Kishi et al.	N/A	N/A
2016/0087022	12/2015	Tsai et al.	N/A	N/A
2016/0171928	12/2015	Gai et al.	N/A	N/A
2017/0011005	12/2016	Tinker	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101063758	12/2006	CN	N/A
101354864	12/2008	CN	N/A
101449311	12/2008	CN	N/A
101958339	12/2010	CN	N/A
102222468	12/2010	CN	N/A
102854665	12/2012	CN	N/A
103000632	12/2012	CN	N/A
103295962	12/2012	CN	N/A
103839517	12/2013	CN	N/A
104064149	12/2013	CN	N/A
104282266	12/2014	CN	N/A

104332485	12/2014	CN	N/A
204167325	12/2014	CN	N/A
1640765	12/2004	EP	N/A
2626851	12/2012	EP	N/A
2743990	12/2013	EP	N/A
2000-284722	12/1999	JP	N/A
2003173154	12/2002	JP	N/A
2004177796	12/2003	JP	N/A
2005195756	12/2004	JP	N/A
2007286150	12/2006	JP	N/A
2007298649	12/2006	JP	N/A
2008165028	12/2007	JP	N/A
2009025735	12/2008	JP	N/A
210-3910	12/2009	JP	N/A
2010003910	12/2009	JP	N/A
2010-056356	12/2009	JP	N/A
2010-224403	12/2009	JP	N/A
2010256914	12/2009	JP	N/A
2011048339	12/2010	JP	N/A
2011054936	12/2010	JP	N/A
2011054951	12/2010	JP	N/A
2011076078	12/2010	JP	N/A
2011107685	12/2010	JP	N/A
2011141529	12/2010	JP	N/A
2011209405	12/2010	JP	N/A
2012093707	12/2011	JP	N/A
2012098316	12/2011	JP	N/A
2012186083	12/2011	JP	N/A
2009-122253	12/2011	JP	N/A
2012-255840	12/2011	JP	N/A
2013-051423	12/2012	JP	N/A
2013047802	12/2012	JP	N/A
2013161084	12/2012	JP	N/A
2014106539	12/2013	JP	N/A
2015079241	12/2014	JP	N/A
2015111703	12/2014	JP	N/A
2015138154	12/2014	JP	N/A
2015156486	12/2014	JP	N/A
2015225104	12/2014	JP	N/A
10-2007-0078470	12/2006	KR	N/A
10-2009-0085231	12/2008	KR	N/A
10-2010-0031342	12/2009	KR	N/A
10-2010-0083322	12/2009	KR	N/A
10-2010-0086256	12/2009	KR	N/A
10-2011-0015380	12/2010	KR	N/A
10-2011-0132814	12/2010	KR	N/A
10-2011-0133281	12/2010	KR	N/A
10-2008-0002237	12/2012	KR	N/A
10-20130074147	12/2012	KR	N/A
20150098281	12/2014	KR	N/A

20150100515	12/2014	KR	N/A
2011125353	12/2010	WO	N/A
2012046658	12/2011	WO	N/A
2012176422	12/2011	WO	N/A
2015052991	12/2014	WO	N/A

OTHER PUBLICATIONS

Yueyue, “Principle and Application of Digital Television Technology”, Beijing Youdian Xueyuan Publishing House, Jun. 1, 2013, 27 pages. cited by applicant

Zhou, “Pacing to the world of LED”, Pacing to the world of LED-Elementary Chapters, pp. 93-98, Jul. 1, 2012, National Defense Industry Publication House, ISBN No. 9787118082081. cited by applicant

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Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 18/532,835, filed Dec. 7, 2023, which is a continuation of U.S. patent application Ser. No. 18/172,049, filed Feb. 21, 2023, now U.S. Pat. No. 11,875,745, which is a continuation of U.S. patent application Ser. No. 17/884,297, filed Aug. 9, 2022, now U.S. Pat. No. 11,615,746, which is a continuation of U.S. patent application Ser. No. 17/555,694, filed Dec. 20, 2021, now U.S. Pat. No. 11,462,163, which is a continuation of U.S. patent application Ser. No. 17/222,844, filed Apr. 5, 2021, now U.S. Pat. No. 11,232,748, which is a continuation of U.S. patent application Ser. No. 16/897,129, filed Jun. 9, 2020, now U.S. Pat. No. 10,997,917, which is a continuation of U.S. patent application Ser. No. 16/678,148, filed Nov. 8, 2019, now U.S. Pat. No. 10,714,009, which is a continuation of U.S. patent application Ser. No. 16/289,007, filed Feb. 28, 2019, now U.S. Pat. No. 10,504,432, which is a continuation of U.S. patent application Ser. No. 15/811,406, filed Nov. 13, 2017, now U.S. Pat. No. 10,354,585, which is a continuation of U.S. patent application Ser. No. 15/263,803, filed Sep. 13, 2016, now U.S. Pat. No. 9,818,344, which claims the benefit of U.S. provisional patent application No. 62/263,074, filed Dec. 4, 2015, which are hereby incorporated by reference herein in their entireties. This application claims the benefit of and claims priority to U.S. patent application Ser. No. 18/532,835, filed Dec. 7, 2023, U.S. patent application Ser. No. 18/172,049, filed Feb. 21, 2023, now U.S. Pat. No. 11,875,745, U.S. patent application Ser. No. 17/884,297, filed Aug. 9, 2022, now U.S. Pat. No. 11,615,746, U.S. patent application Ser. No. 17/555,694, filed Dec. 20, 2021, now U.S. Pat. No. 11,462,163, U.S. patent application Ser. No. 17/222,844, filed Apr. 5, 2021, now U.S. Pat. No. 11,232,748, U.S. patent application Ser. No. 16/897,129, filed Jun. 9, 2020, now U.S. Pat. No. 10,997,917, U.S. patent application Ser. No. 16/678,148, filed Nov. 8, 2019, now U.S. Pat. No. 10,714,009, U.S. patent application Ser. No. 16/289,007, filed Feb. 28, 2019, now U.S. Pat. No. 10,504,432, U.S. patent application Ser. No. 15/811,406, filed Nov. 13, 2017, now U.S. Pat. No. 10,354,585, U.S. patent application Ser. No. 15/263,803, filed Sep. 13, 2016, now U.S. Pat. No. 9,818,344, and U.S. provisional patent application No. 62/263,074, filed Dec. 4, 2015.

BACKGROUND

(1) This relates generally to electronic devices, and, more particularly, to electronic devices with displays.

(2) Electronic devices often include displays. Displays such as organic light-emitting diode displays have pixels with light-emitting diodes.

(3) It can be challenging to design displays with light-emitting diodes. If care is not taken, high transistor leakage currents, slow transistor switching speeds, routing complexity, voltage drops due to ohmic losses, and other issues may adversely affect display performance.

SUMMARY

(4) An electronic device may have a display. The display may have an array of pixels organized in rows and columns. Each of the pixels may have a light-emitting diode such as an organic light-emitting diode that emits light in response to application of a drive current. A drive transistor in each pixel may supply the drive current to the light-emitting diode of that pixel in response to a gate-source voltage across a gate and source of the drive transistor.

(5) The source of each drive transistor may be coupled to a positive power supply. An emission transistor may be coupled in series with the drive transistor and the light-emitting diode of each pixel between the positive power supply and a ground power supply. The pixels may include first and second switching transistors. A data storage capacitor may be coupled between the gate and the source of the drive transistor in each pixel. Control signals may be provided to gates of the switching transistors and the emission transistor from display driver circuitry.

(6) Signal lines may be provided in columns of pixels to route signals such as data signals, sensed drive currents from the drive transistors, and predetermined voltages such as reference voltages between the display driver circuitry and the pixels. The switching transistors, emission transistors, and drive transistors may include semiconducting-oxide transistors and silicon transistors and may be n-channel transistors or p-channel transistors.

(7) Further features will be more apparent from the accompanying drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an illustrative display in accordance with an embodiment.

(2) FIG. 2 is a circuit diagram of an illustrative pixel for a display in accordance with an embodiment.

(3) FIGS. 3 and 4 are timing diagrams showing illustrative signals involved in operating a display with pixels of the type shown in FIG. 2 in accordance with an embodiment.

(4) FIG. 5 is a circuit diagram of another illustrative pixel for a display in accordance with an embodiment.

(5) FIGS. 6 and 7 are timing diagrams showing illustrative signals involved in operating a display with pixels of the type shown in FIG. 5 in accordance with an embodiment.

(6) FIG. 8 is a circuit diagram of an additional illustrative pixel for a display in accordance with an embodiment.

(7) FIGS. 9 and 10 are timing diagrams showing illustrative signals involved in operating a display with pixels of the type shown in FIG. 8 in accordance with an embodiment.

(8) FIG. 11 is a circuit diagram of a further illustrative pixel for a display in accordance with an embodiment.

(9) FIGS. 12 and 13 are timing diagrams showing illustrative signals involved in operating a display with pixels of the type shown in FIG. 11 in accordance with an embodiment.

(10) FIG. 14 is a diagram of an illustrative pixel circuit with five transistors and one capacitor in accordance with an embodiment.

(11) FIG. 15 is a timing diagram showing signals involved in operating a display with pixels of the type shown in FIG. 14 in accordance with an embodiment.

- (12) FIG. **16** is a diagram of the pixel circuit of FIG. **14** during on-bias stress operations in accordance with an embodiment.
- (13) FIG. **17** is a diagram of the pixel circuit of FIG. **14** during data writing operations in accordance with an embodiment.
- (14) FIG. **18** is a diagram of the pixel circuit of FIG. **14** during emission operations in accordance with an embodiment.
- (15) FIG. **19** is diagram of the pixel circuit of FIG. **14** when gathering threshold voltage information in accordance with an embodiment.
- (16) FIGS. **20A** and **20B** are timing diagrams showing signals involved in operating a display with pixels as shown in FIG. **14** in accordance with an embodiment.
- (17) FIG. **21** is a diagram of the pixel circuit of FIG. **14** when gathering threshold voltage information in accordance with another embodiment.
- (18) FIG. **22** is a timing diagram showing signals involved in operating a display with pixels as shown in FIG. **21** in accordance with an embodiment.
- (19) FIG. **23** is a circuit diagram of an illustrative pixel with a bypass transistor in accordance with an embodiment.
- (20) FIG. **24** is a diagram showing control signals of the type that may be used in operating the pixel of FIG. **23** in accordance with an embodiment.
- (21) FIG. **25** is a circuit diagram of another illustrative pixel with a bypass transistor in accordance with an embodiment.
- (22) FIG. **26** is a diagram showing control signals of the type that may be used in operating the pixel of FIG. **25** in accordance with an embodiment.
- (23) FIGS. **27**, **28**, **29**, **30**, and **31** show illustrative operations for a pixel of the type shown in FIG. **25**.
- (24) FIG. **32** is a diagram showing how current sensing operations of the type described in connection with FIG. **30** may be performed.

DETAILED DESCRIPTION

- (25) Displays such as display **14** of FIG. **1** may be used in devices such as tablet computers, laptop computers, desktop computers, displays, cellular telephones, media players, wristwatch devices or other wearable electronic equipment, or other suitable electronic devices.
- (26) Display **14** may be an organic light-emitting diode display or may be a display based on other types of display technology (e.g., displays with light-emitting diodes formed from discrete crystalline semiconductor dies, displays with quantum dot light-emitting diodes, etc.). Configurations in which display **14** is an organic light-emitting diode display are sometimes described herein as an example. This is, however, merely illustrative. Any suitable type of display may be used, if desired.
- (27) Display **14** may have a rectangular shape (i.e., display **14** may have a rectangular footprint and a rectangular peripheral edge that runs around the rectangular footprint) or may have other suitable shapes. Display **14** may be planar or may have a curved profile.
- (28) As shown in FIG. **2**, display **14** may have an array of pixels **22** formed on substrate **24**. Substrate **24** may be formed from glass, metal, plastic, ceramic, or other substrate materials. Pixels **22** may receive data signals and other signals over paths such as vertical paths **16**. Each vertical path **16** may be associated with a respective column of pixels **22** and may contain one or more signal lines. Pixels **22** may receive horizontal control signals (sometimes referred to as emission enable control signals or emission signals, scan signals, or gate signals) over paths such as horizontal paths **18**. Each horizontal path **18** may contain one or more horizontal signal lines.
- (29) There may be any suitable number of rows and columns of pixels **22** in display **14** (e.g., tens or more, hundreds or more, or thousands or more). Each pixel **22** may have a light-emitting diode that emits light under the control of a pixel circuit formed from thin-film transistor circuitry (e.g., thin-film transistors, thin-film capacitors, etc.). The thin-film transistor circuitry of pixels **22** may

include silicon thin-film transistors such as polysilicon thin-film transistors, semiconducting-oxide thin-film transistors such as indium gallium zinc oxide transistors, or thin-film transistors formed from other semiconductors. Pixels **22** may contain light-emitting diodes of different colors (e.g., red, green, and blue diodes for red, green, and blue pixels, respectively) to provide display **14** with the ability to display color images.

(30) Pixels **22** may be arranged in a rectangular array or an array of other shapes. The array of pixels **22** forms an active area for display **14** and is used in displaying images for a user. Inactive portions of display **14** may run along one or more of the edges of active area AA. Inactive areas form borders for display **14** and may be free of pixels **22**.

(31) Display driver circuitry **20** may be used to control the operation of pixels **22**. Display driver circuitry **20** may be formed from integrated circuits, thin-film transistor circuits, or other suitable circuitry and may be located in the inactive area of display **14**. Display driver circuitry **20** may contain communications circuitry for communicating with system control circuitry such as a microprocessor, storage, and other storage and processing circuitry. During operation, the system control circuitry may supply circuitry **20** with information on images to be displayed on display **14**.

(32) To display the images on pixels **22**, display driver circuitry such as circuitry **20A** may supply image data to vertical lines **16** while issuing clock signals and other control signals to supporting display driver circuitry such as display driver circuitry **20B** (e.g., gate driver circuitry) over path **26**. If desired, circuitry **20** may also supply clock signals and other control signals to gate driver circuitry **20B** on an opposing edge of display **14**.

(33) Gate driver circuitry **20B** (sometimes referred to as horizontal control line control circuitry) may be implemented as part of an integrated circuit and/or may be implemented using thin-film transistor circuitry. Horizontal control lines **18** in display **14** may carry gate line signals (e.g., scan line signals, emission enable control signals, and other horizontal control signals) for controlling the pixels of each row. There may be any suitable number of horizontal control signals per row of pixels **22** (e.g., one or more, two or more, three or more, four or more, etc.).

(34) Pixels **22** may each include a drive transistor coupled in series with a light-emitting diode. An emission enable transistor (emission transistor) may be coupled in series with the drive transistor and light-emitting diode between positive and ground power supply terminals. A storage capacitor in each pixel may be used to store loaded data (e.g., data establishing a pixel brightness value for the pixel) between successive image frames. Each pixel may also have one or more switching transistors to support data loading operations and other operations.

(35) The frame rate of display **14** may be 60 Hz or other suitable frame rate. If desired, display **14** may support variable refresh rate operations. During normal refresh rate operations, the refresh rate of display **14** may be relatively high (e.g., 60 Hz). When static content is being displayed on display **14**, the refresh rate of display **14** may be lowered (e.g., to 1-5 Hz or other suitable low refresh rate) to conserve power.

(36) The circuitry of pixels **22** (e.g., transistors such as drive transistors, light-emitting diodes, etc.) may be influenced by aging effects. Display driver circuitry **20** (e.g., circuitry **20A**) may contain current sensing circuitry and other compensation circuitry that periodically measures the performance of pixels **22**. Based on these periodic measurements (e.g., periodic current sensing measurements to measure the current produced by the drive transistors of the pixels), display driver circuitry **20** may make adjustments to the data that is loaded into pixels **22**. The adjustments that are made to the loaded pixel data may compensate for measured pixel performance variations (e.g., the adjustments may compensate for aging effects, thereby ensuring that display **14** exhibits a desired uniformity and other attributes). Current sensing (e.g., sensing of the current of drive transistors in pixels **22**) may be performed using vertical lines in display **14** such as lines **16**. During normal operation (sometimes referred to as the “emission” mode of display **14**), emission control lines can be asserted to turn on the emission enable transistors in pixels **22**. The emission enable transistors may be turned off during data loading and current sensing operations.

(37) Pixels **22** may use both semiconducting-oxide transistors and silicon transistors. Semiconducting-oxide transistors tend to exhibit lower leakage current than silicon transistors. Silicon transistors tend to switch more quickly than semiconducting-oxide transistors. By appropriate selection of which transistors in each pixel are semiconducting-oxide transistors and which transistors in each pixel are silicon transistors and by configuring the horizontal lines, vertical lines, and other pixel circuitry appropriately, display performance can be optimized. FIGS. **2-13** show various pixel circuit arrangements and associated signal timing diagrams associated with illustrative embodiments for display **14**.

(38) As shown in the illustrative configuration for pixel **22** of FIG. **2**, each pixel **22** may contain a light-emitting diode such as light-emitting diode **30** that emits light **32** in response to application of drive current I_d . Light-emitting diode **30** may be, for example, an organic light-emitting diode. The transistors and capacitor structures of pixels **22** may be formed from thin-film circuitry on substrate **24** (FIG. **1**). In general, each pixel **22** of display **14** may include p-channel transistors, n-channel transistors, semiconducting-oxide transistors, silicon transistors, one or more storage capacitors, and signal paths (e.g., portions of one or more vertical signal lines, and one or more horizontal signal lines).

(39) In the example of FIG. **2**, light-emitting diode **30** is coupled in series with emission enable transistor (emission transistor) TE and drive transistor TD between positive power supply V_{ddel} and ground power supply V_{ssel} . Storage capacitor Cst1 maintains a loaded data value on Node2, which is connected to the gate of drive transistor TD. Source S of drive transistor TD is coupled to positive power supply V_{ddel} . The value of the gate-source voltage V_{gs} of drive transistor TD (i.e., the voltage difference between Node2 and power supply terminal V_{ddel} at source S of transistor TD) establishes the drive current I_d through light-emitting diode **30**. Emission is enabled or disabled using emission control signal EM, which is applied to the gate of emission transistor TE. Switching transistors T1 and T2 are used for data loading and current sensing operations. Transistors T1, T2, TD, and TE may all be p-channel silicon transistors (as an example).

(40) Each column of pixels **22** such as pixel **22** of FIG. **2** may be associated with a pair of vertical signal lines **16**. The vertical signal lines may include a data line (Data) and a reference voltage line (V_{ref}). The data line may be used to load data onto data storage capacitor Cst1. The reference voltage line, which may sometimes be referred to as a sense line, may be used to measure the current of drive transistor TD (e.g., to assess aging) during current sensing operations. The reference voltage line may also be used in loading predetermined voltages onto a node between emission transistor TE and light-emitting diode **30** (i.e., Node3).

(41) Each row of pixels **22** such as pixel **22** of FIG. **2** may be associated with three horizontal signal lines **18**. The horizontal signal lines **18** may include a first switching transistor control signal (scan signal) Scan1 that is applied to the gate of switching transistor T1, a second switching transistor control signal (scan signal) Scan2 that is applied to the gate of switching transistor T2, and an emission enable signal (emission signal) EM that is applied to the gate of emission transistor TE.

(42) A signal timing diagram showing signals associated with loading data from data line Data onto storage capacitor Cst1 at Node2 of pixel **22** of FIG. **2** is shown in FIG. **3**. During normal operation (emission operations), EM is held low by display driver circuitry **20B**, so transistor TE is on. With TE on, the data value on node Node2 establishes a desired V_{gs} value across gate G and source S of drive transistor TD (Source S is tied to V_{ddel}), thereby setting the magnitude of drive current I_d for light-emitting diode **30**. During data loading operations, EM is taken high by circuitry **20B** to turn off transistor TE and block current I_d . While EM is high, circuitry **20B** takes signals Scan1 and Scan2 low to turn on transistors T1 and T2. With T2 on, a known reference voltage may be supplied to Node3 from line V_{ref} . With T1 on, the current data signal on the data line (Data) may be loaded onto capacitor Cst1 at Node2. Emission operations may then be resumed by taking EM low and taking Scan1 and Scan2 high. During emission, the data value loaded onto capacitor Cst1

at Node2 determines the output level of light 32 from light-emitting diode 30.

(43) A signal timing diagram showing signals associated with current sensing operations (which may be performed periodically such as once per hour, once per week, etc. by interrupting normal emission operations) is shown in FIG. 4.

(44) During preloading, EM is taken high to prevent current from flowing through light-emitting diode 30 while Scan1 and Scan2 are taken low. While Scan2 is low, transistor T2 is turned on and a known reference voltage is loaded onto Node3 from line Vref. While Scan1 is low, known reference data ("sense data") is loaded from line Data onto Node2, via transistor T1, which is on. This establishes known conditions for operating drive transistor TD (e.g., a predetermined Vgs value and predetermined voltage on Node3).

(45) After loading pixel 22 with sense data, current sensing operations are performed. During sensing operations, EM is taken low and Scan2 is held low while Scan1 is taken high. This routes the current that is flowing through drive transistor TD into line Vref, which then serves as a sense line. Current sensing circuitry within the compensation circuits of display driver circuitry 20B measures the amount of current flowing through transistor TD so that the performance of transistor TD may be assessed. The compensation circuitry of display driver circuitry 20B can use current measurements such as these to compensate pixels 22 for aging effects (e.g., aging that affects the amount of drive current Id that transistor TD produces for a given Vgs value).

(46) After current sensing operations are complete, data may be loaded from data line Data onto Node2 by taking EM high, taking Scan1 low to turn on transistor T1, and holding Scan2 low. Pixel 22 may be placed in emission mode after data has been loaded by taking EM low to turn on transistor TE and taking Scan1 and Scan2 high to turn off transistors T1 and T2.

(47) The configuration for pixel 22 of FIG. 2 uses three gate control signals on three horizontal control lines in a row of pixels 22 and routes data, reference voltage signals, and current measurements over two vertical lines in each column of pixels 22. The vertical lines of each column operate independently of the vertical lines of the other columns (i.e., there are N independent lines Data and N independent lines Vref in a display having N columns of pixels 22).

(48) To reduce transistor leakage current and thereby allow display 14 to be operated efficiently at a low refresh rate (e.g., when display 14 is configured to support variable refresh rate operation), pixel 22 may be provided with a semiconducting-oxide switching transistor. For example, data loading transistor T1 of pixel 22 of FIG. 5 may be an n-channel semiconducting-oxide transistor. Transistors TE, TD, and T2 may be p-channel silicon transistors.

(49) A signal timing diagram showing signals associated with loading data from data line Data onto storage capacitor Cst1 at Node2 in pixel 22 of FIG. 5 is shown in FIG. 6.

(50) During normal operation (emission operations) of pixel 22 of FIG. 5, EM is held low by display driver circuitry 20B, so transistor TE is on. Source S of drive transistor TD is at Vddel. With TE on, the data value on node Node2 establishes a desired gate-source voltage Vgs value across gate G and source S of drive transistor TD, thereby setting the magnitude of drive current Id for light-emitting diode 30.

(51) During data loading operations, EM is taken high by circuitry 20B to turn off transistor TE and block current Id. While EM is high, circuitry 20B takes signal Scan1 high and takes Scan2 low to turn on transistors T1 and T2. Transistor T1 is a semiconducting-oxide transistor, so it may be desirable to extend the amount of time that Scan1 is high (relative to a scenario in which T1 is a silicon transistor) to ensure sufficient time for the transistor T1 to settle. With T2 on for data loading, a known reference voltage may be supplied to Node3 from line Vref. With T1 on, the data signal that is present on the data line (Data) may be loaded onto capacitor Cst1 at Node2. Emission operations may then be resumed by taking EM and Scan low and taking Scan2 high.

(52) A signal timing diagram showing signals associated with periodic current sensing operations for pixel 22 of FIG. 5 is shown in FIG. 7.

(53) During preloading of pixel 22 of FIG. 5, EM is taken high to prevent current from flowing

through light-emitting diode **30**, while Scan1 is taken high and Scan2 is taken low. With Scan2 low, transistor T2 is turned on and a known reference voltage is loaded onto Node3 from line Vref. With Scan1 high, known reference data ("sense data") is loaded from line Data onto Node2, via transistor T1, which is on. This establishes known conditions for operating drive transistor TD (e.g., a predetermined Vgs value and predetermined voltage on Node3).

(54) During sensing operations for pixel **22** of FIG. 5, EM and Scan1 are taken low and Scan2 is held low. This routes the current that is flowing through drive transistor TD into line Vref, which serves as a sense line. Current sensing circuitry within the compensation circuits of display driver circuitry **20B** measures the amount of current flowing through transistor TD so that the performance of transistor TD may be assessed. As with the scenario of FIG. 2, the compensation circuitry of display driver circuitry **20B** can use current measurements such as these to compensate pixels **22** of FIG. 5 for aging effects (e.g., aging that affects the amount of drive current Id that transistor TD produces for a given Vgs value).

(55) After sensing operations are complete, data may be loaded from data line Data onto Node2 by taking EM and Scan1 high while holding Scan2 low. Pixel **22** may be placed in emission mode after data has been loaded by taking EM and Scan1 low and taking Scan2 high, thereby turning on transistor TE and turning off transistors T1 and T2.

(56) Because the EM and Scan1 signals are identical, the functions of these signals can be implemented using a single combined signal that is carried on a single signal lines (i.e., a single signal EM/Scan1 can replace the separately adjusted EM and Scan1 signals of pixel **22** of FIG. 2). The configuration for pixel **22** of FIG. 5 therefore uses only two gate control signals on two horizontal control lines, saving routing resources. Two vertical lines (Data and Vref) may be used to carry data, reference voltage signals, and current measurements in each column of pixels **22**. The vertical lines of each column of a display with pixels **22** of the type shown in FIG. 5 operate independently of the vertical lines of the other columns (i.e., there are N independent lines Data and N independent lines Vref in a display having N columns of pixels **22**).

(57) If desired, the number of horizontal control signals that are associated with each row of pixels **22** can be reduced further using circuitry of the type shown in pixel **22** of FIG. 8. In the configuration of FIG. 8, transistors T1 and T2 are both n-channel semiconducting-oxide transistors, whereas transistors TE and TD are both p-channel silicon transistors. The use of semiconducting-oxide transistors in pixel **22** (e.g., for transistors T1) helps to reduce leakage current and thereby allow display **14** to be operated efficiently at a low refresh rate (e.g., when display **14** is configured to support variable refresh rate operation).

(58) A signal timing diagram showing signals associated with loading data from data line Data onto storage capacitor Cst1 at Node2 in pixel **22** of FIG. 8 is shown in FIG. 9.

(59) During normal operation (emission operations) of pixel **22** of FIG. 8, EM is held low by display driver circuitry **20B**, so transistor TE is on. With TE on, the data value on node Node2 establishes a desired Vgs value across gate G and source S of drive transistor TD, thereby setting the magnitude of drive current Id for light-emitting diode **30**. Signals Scan1 and Scan2 may be held low during emission to turn off transistors T1 and T2 during emission.

(60) During data loading operations, EM is taken high by circuitry **20B** to turn off transistor TE and block current Id. While EM is high, circuitry **20B** takes signals Scan1 and Scan2 high to turn on transistors T1 and T2. Transistor T1 is a semiconducting-oxide transistor, so it may be desirable to extend the amount of time that Scan1 is high (relative to a scenario in which T1 is a silicon transistor) to ensure sufficient time for the transistor T1 to settle. With T2 on for data loading, a known reference voltage may be supplied to Node3 between transistor TE and light-emitting diode **30** from line Vref. With T1 on, the data signal that is present on the data line (Data) may be loaded onto capacitor Cst1 at Node2. Emission operations may then be resumed by taking EM, Scan1, and Scan2 low.

(61) A signal timing diagram showing signals associated with periodic current sensing operations

for pixel **22** of FIG. **8** is shown in FIG. **10**.

(62) During preloading of pixel **22** of FIG. **8**, EM is taken high to prevent current from flowing through light-emitting diode **30**, while Scan1 and Scan2 are taken high. With Scan2 high, transistor T2 is turned on and a known reference voltage is loaded onto Node3 from line Vref. With Scan1 high, known reference data ("sense data") is loaded from line Data onto Node2, via transistor T1, which is on. This establishes known conditions for operating drive transistor TD (e.g., a predetermined Vgs value and predetermined voltage on Node3).

(63) During sensing operations for pixel **22** of FIG. **8**, EM and Scan1 are taken low and Scan2 is held high. This routes the current that is flowing through drive transistor TD into sense line Vref. Current sensing circuitry within the compensation circuits of display driver circuitry **20B** measures the amount of current flowing through transistor TD so that the performance of transistor TD may be assessed. As with the scenario of FIG. **2**, the compensation circuitry of display driver circuitry **20B** can use current measurements such as these to compensate pixels **22** of FIG. **8** for aging effects (e.g., aging that affects the amount of drive current Id that transistor TD produces for a given Vgs value).

(64) After sensing operations are complete, data may be loaded from data line Data onto Node2 by taking EM and Scan1 high while holding Scan2 high. Pixel **22** may be placed in emission mode after data has been loaded by taking EM, Scan1, and Scan2 low, thereby turning on transistor TE and turning off transistors T1 and T2.

(65) Because the EM, Scan1, and Scan2 signals are identical (i.e., because transistor T2 is an n-channel transistor like transistor T1), the functions of these signals can be implemented using a single combined signal that is carried on a single signal line (i.e., a single signal EM/Scan1/Scan2 can replace the separately adjusted EM, Scan1, and Scan2 signals of pixel **22** of FIG. **2**). The configuration for pixel **22** of FIG. **5** therefore uses only a single gate control signal on a single associated horizontal control line in each row of pixels **22**, which helps to minimize routing resources. Two vertical lines (Data and Vref) may be used to carry data, reference voltage signals, and current measurements in each column of pixels **22**. The vertical lines of each column of a display with pixels **22** of the type shown in FIG. **8** operate independently of the vertical lines of the other columns (i.e., there are N independent lines Data and N independent lines Vref in a display having N columns of pixels **22**).

(66) Pixels with configurations of the type shown in FIGS. **2**, **5**, and **8** may be sensitive to variations in Vddel that arise from IR drops (ohmic losses) as Vddel is distributed across display **14**. This is because the source voltage at the source S of drive transistor TD is coupled to Vddel and can vary as Vddel varies due to the position of each pixel **22** within display **14**.

(67) If desired, a pixel circuit of the type shown in FIG. **11** may be used for pixels **22** to help reduce performance variations due to Vddel variations. In the illustrative configuration of FIG. **11**, T1 is coupled between line Vref and Node2, whereas transistor T2 is coupled between data line Data and Node1. Transistor T2 may therefore serve as a data loading transistor. Node2 is coupled to the gate of drive transistor TD.

(68) During emission operations, the voltage on capacitor Cst1 (i.e., the voltage on Node2) is preferably maintained at a constant level to ensure a steady output level for light **32**. During operations such as variable refresh rates operations, the refresh rate of display **14** may be relatively low (e.g., 1-5 Hz). To prevent transistor leakage current that might adversely affect the stability of the data voltage at Node2, transistor T1 may be implemented using a semiconducting-oxide transistor (e.g., a n-channel semiconducting-oxide transistor). Transistors TE, TD, and T2 may be p-channel silicon transistors. Because transistor T2 is a silicon transistor, data may be rapidly loaded from data line Data to Node1.

(69) Unlike the arrangements of FIGS. **2**, **5**, and **8**, source S of drive transistor TD of FIG. **11** is connected to Node1, rather than Vddel. The level of voltage Vddel may vary due to IR losses as Vddel is distributed across display **14**, but the voltage Vs on source S will not vary across display

14 (i.e., V_s will be independent of the position of pixel **22** within display **14**) because the voltage V_s is established by loading a predetermined reference voltage onto Node1 via transistor T2 from data line Data.

(70) A signal timing diagram showing signals associated with loading data from data line Data onto storage capacitor Cst1 at Node1 of pixel **22** of FIG. **11** is shown in FIG. **12**.

(71) During normal operation (emission operations), EM is held low by display driver circuitry **20B**, so transistor TE is on. Scan1 is low to maintain transistor T1 in an off state. Scan2 is high to maintain transistor T2 in an off state. With TE on, the data value on node Node1 (and the voltage on Node2) establishes a desired V_{gs} value across gate G and source S of drive transistor TD, thereby setting the magnitude of drive current I_d for light-emitting diode **30**.

(72) During data loading operations, EM is taken high by circuitry **20B** to turn off transistor TE and block current I_d . While EM is high, circuitry **20B** takes signal Scan1 high to turn transistor T1 on. With transistor T1 on, Node2 is precharged to a predetermined voltage, thereby establishing a known gate voltage V_g at Node2 of transistor TD. Scan2 is initially high, which holds T2 off. When Scan2 is taken low (which may take place one row time before emission starts, two row times before emission starts, or at any other suitable time), transistor T2 is turned on and a desired data value is loaded from data line Data to Node1 via transistor T2. Emission operations may then be resumed by taking EM low, taking Scan1 low, and taking Scan2 high.

(73) A signal timing diagram showing signals associated with periodic current sensing operations for pixel **22** of FIG. **11** is shown in FIG. **13**.

(74) During preloading, EM is taken high to prevent current from flowing through light-emitting diode **30** while Scan1 is taken high and Scan2 is taken low. With Scan2 low, transistor T2 is turned on and known reference data ("sense data") is loaded from line Data onto Node1. With Scan1 high, transistor T1 is turned on and a predetermined voltage (e.g., $-5.5V$ or other suitable value) is provided from reference voltage line V_{ref} to Node2. This establishes known conditions for operating drive transistor TD (e.g., a predetermined V_{gs} value).

(75) During sensing operations, EM is held high, Scan1 is taken low, and Scan2 is held low. This holds TE off, turns off T1, and holds T2 on, thereby routing the current that is flowing through drive transistor TD through line Data, which is therefore serving as a sense line. Current sensing circuitry within the compensation circuits of display driver circuitry **20B** measures the amount of current flowing through transistor TD via line Data, so that the performance of transistor TD may be assessed. Current sensing may take place over a time period of 100 microseconds or other suitable time period. The compensation circuitry of display driver circuitry **20B** can use current measurements such as these to compensate pixels **22** for aging effects (e.g., aging that affects the amount of drive current I_d that transistor TD produces for a given V_{gs} value).

(76) After current sensing operations are complete, data may be loaded into pixel **22** by holding EM high to turn off transistor TE, by taking Scan1 high to turn on transistor T1 and thereby transfer a predetermined voltage from V_{ref} to Node2, and by holding Scan2 low to hold transistor T2 on so that a desired data signal passes from data line Data to Node1. Pixel **22** may be placed in emission mode after data has been loaded by taking EM low to turn on transistor TE, taking Scan1 low to turn off transistor T1, and taking Scan2 high to turn off transistor T2.

(77) The voltage range of signal EM may be $-10V$ to $8V$, may be $-8V$ to $8V$, or may be any other suitable voltage range. The voltage of V_{ddel} may be $5-8V$ or other suitable positive power supply voltage level. The voltage of V_{ssel} may be $-2V$ or other suitable ground power supply voltage level. The voltage range of the signals on line Data may be $-4.5V$ to $-0.3V$ or other suitable voltage range. The voltage range of Scan2 may be $-10V$ to $-8V$, may be $-12V$ to $-4V$, or may be other suitable voltage range. The voltage range of Scan1 may be $-10V$ to $-8V$, may be $-8V$ to $8V$, or may be other suitable voltage range.

(78) The configuration for pixel **22** of FIG. **2** uses three gate control signals (EM, Scan1, and Scan2) on three horizontal control lines in each row of pixels **22** and routes data, reference voltage

signals, and current measurements using two vertical lines: Vref and Data in each column of pixels **22**. One of the vertical lines (line Data) is a shared line that is used both for current sensing operations and for data loading operations. There is preferably a separate Data line in each column of pixels **22** in display **14**. The other of the vertical lines (line Vref) associated with pixels **22** is part of a global path that may be used to distribute a shared voltage to all of pixels **22** in display **14** in parallel. Because Vref is a global signal path, only a single Vref signal needs be provided by display driver circuitry **20A** to pixels **22** (i.e., there is a reduced need for signal routing resources between display driver circuitry **20B** and pixels **22** compared to scenarios in which separate Vref signal lines are used for respective columns). Only one individual vertical signal line Data need be provided in each column, rather than the two individual vertical signal lines used in arrangements of the type shown in FIGS. **2**, **5**, and **8**. The arrangement of FIG. **11** therefore exhibits low display driver circuitry fan out.

(79) Due to the use of a low-leakage current semiconducting-oxide transistor for transistor **T1**, the refresh rate of display **14** may be lowered to a low rate (e.g., 1-5 Hz) during variable refresh rate operations. Charging times (i.e., the amount of time associated with charging Node**1** to a desired value during data loading operations) may be minimized by using a silicon transistor to implement transistor **T2**. The pixel arrangement of FIG. **11** is also insensitive to variations in Vddel (e.g., variations due to IR drops), because both Node**1** and Node**2** are actively loaded with desired voltages during data loading, thereby establishing a desired gate-source voltage across drive transistor **TD** without using Vddel.

(80) FIG. **14** is a diagram of an illustrative pixel circuit with five transistors and one capacitor. Drive transistor **TD** is coupled in series with emission enable transistors **TE1** and **TE2** and with light-emitting diode **44** (e.g., an organic light-emitting diode) between positive power supply terminal **40** and ground power supply terminal **42**. Horizontal control signals (gate signals) such as emission enable control signals **EM1** and **EM2** may be used to control transistors **TE1** and **TE2**, respectively. Horizontal control signals (gate signals) such as scan control signals **SCAN1** and **SCAN2** may be used to control switching transistors **TS1** and **TS2**, respectively. Transistor **TS1** may be, for example, a semiconducting-oxide transistor and transistors **TS2**, **TE1**, **TE2**, and **TD** may be silicon transistors (as an example). Capacitor **Cst1** may be coupled between Node**2** (at the gate of drive transistor **TD**) and Node**1** (at the source of transistor **TD**). The line Vref may be used to supply a reference voltage to a column of pixels **22**. Data signals (**D**) may be supplied to pixel **22** using data line **Data**.

(81) FIG. **15** is a timing diagram showing signals involved in operating a display with pixels of the type shown in FIG. **14**. As shown in FIG. **15**, on-bias stress may be applied during the operations of on-bias stress period **200**, data writing may be performed during data writing period **202**, and emission operations may be performed during emission period **204**.

(82) FIG. **16** is a diagram of the pixel circuit of FIG. **14** during on-bias stress period **200**. During this period, transistor **TE2** is turned off to prevent drive current from flowing through diode **44** and transistor **TS1** is turned on to supply on-bias stress to the gate of drive transistor **TD** to precondition transistor **TD**. Voltage V_{gs} of transistors **TD** is high because **TE1** is on and Node**1** is at Vddel and because **TS1** is on and Node**2** is at Vref.

(83) FIG. **17** is a diagram of the pixel circuit of FIG. **14** during data writing operations (period **202** of FIG. **15**). During data writing, transistor **TS1** is initially turned on to load a known reference voltage Vref onto Node**2** while transistor **TS2** is turned on to load a data signal (sometimes referred to as V_{data} , **Data**, or signal **D**) onto Node**1**. Transistors **TE1** is turned off to isolate Node**1** from Vddel. This creates a voltage $V_{data}-V_{ref}$ across capacitor **Cst1**. Transistor **TS1** and transistor **TS2** are then turned off and transistor **TE1** is turned on, as shown in FIG. **18**. With **TE1** on, the voltage at Node**1** is taken to Vddel. The voltage across capacitor **Cst1** does not change instantaneously, so when Node**1** is taken to Vddel, Node**2** is taken to $V_{ddel}-(V_{data}-V_{ref})$. Current flow through diode **44** and therefore light emission **46** is therefore proportional to V_{data} during emission period **204**.

(84) FIGS. **19**, **20A**, **20B**, **21**, and **22** illustrate how display driver circuitry **20** may compensate display **14** for variations in the threshold voltage V_t of drive transistors such as transistor TD in pixels **22** of display **14**.

(85) FIG. **19** is diagram of the pixel circuit of FIG. **14** when gathering threshold voltage information in accordance with an arrangement of the type that may sometimes be referred to as a “current sensing” arrangement. FIG. **20A** is a timing diagram showing signals involved in operating gathering the threshold voltage information. As shown in FIG. **20A**, on-bias stress may be applied to transistor TD during on-bias stress period **200**. During period **202'**, predefined data for use during threshold voltage compensation operations may be loaded into pixel **22** (i.e., a known voltage may be applied across capacitor Cst as described in connection with loading Vdata onto Node1 in connection with FIG. **17**). Image data may be loaded into pixel **22** during data writing period **202** and the loaded image data may be used to control the amount of light emitted by diode **44** during emission period **204**. Between periods **202'** and **202**, display driver circuitry **20** may, during sensing period **206**, measure the threshold voltage V_t of drive transistor TD. To determine the threshold voltage V_t of transistor TD, a known reference data value V_{ref} is written during period **202'**. Then current flow on data line Data is measured with a current sensor and threshold voltage V_t is computed from the measured current. During period **202**, data that has been externally compensated for any variations in V_t may then be written into pixel **22**. Each of the pixels **22** in display **14** such as pixel **22** of FIG. **14** can be compensated for any measured variation in threshold voltage V_t by adjusting the value of the image data that display driver circuitry **20** supplies to pixel **22** during period **202** (i.e., display driver circuitry **20** may implement an external threshold voltage compensation scheme).

(86) FIG. **19** shows the operation of pixel **22** during sensing period **206** (sometimes referred to as threshold voltage sensing or current sensing). As shown in FIG. **19**, transistor TE1 is turned off during period **206** to isolate Node1 from Vddel. Transistor TS1 is turned off to allow Node2 to float. During period **206**, the gate-source voltage V_{gs} across transistor TD is determined by the known data loaded into capacitor Cst1 during period **202'**. Transistor TS2 is on, so the known data on transistor TD (and the threshold voltage V_t of transistor TD) determines the current flowing on the Data line. Display driver circuitry **20** measures this current during period **206** to ascertain the value of threshold voltage V_t . Appropriate threshold voltage compensation operations may then be performed by adjusting the values of the image data loaded into pixel **22** during data writing operations **202** (FIG. **20A**).

(87) FIG. **21** is a diagram of the pixel circuit of FIG. **14** when gathering threshold voltage information in accordance with another illustrative external threshold voltage compensation scheme (i.e., a scheme of the type that may sometimes be referred to as a “voltage sensing” scheme). FIG. **22** is a timing diagram showing signals involved in operating a display with pixels as shown in FIG. **21**.

(88) As shown in FIG. **22**, on-bias stress may be applied to transistor TD during on-bias stress period **200**. Image data may be loaded into pixel **22** during data writing period **202** and the loaded image data may be used to control the amount of light emitted by diode **44** during emission period **204**. Between periods **200** and **202**, display driver circuitry **20** may, during sensing period **208**, measure the threshold voltage V_t of drive transistor TD. First transistor TS1 may be turned on to take Node2 to V_{ref} . This establishes a known current on data line Data. Transistors TD and TE2 are on, so current flows through light-emitting diode **44**. The voltage drop across transistors TE2, TD, and TS2 is small, so the resulting voltage V_{oled} on data line Data can be measured. Threshold voltage V_t can then be obtained from the known values of the flowing current and V_{oled} . Pixel **22** can be compensated for any variation in threshold voltage V_t that is measured during sensing period **208** by adjusting the value of the image data that display driver circuitry **20** supplies to pixel **22** during period **202** (i.e., display driver circuitry **20** may implement an external threshold voltage compensation scheme).

(89) FIG. 21 shows the operation of pixel 22 during sensing period 208 (sometimes referred to as voltage sensing or Voled sensing). As shown in FIG. 21, transistor TE1 is turned off during period 208 to isolate Node1 from Vddel. Transistor TS1 is turned on to supply reference voltage Vref to Node2 at gate G of drive transistor TD. A known data voltage Vdata is supplied to Node1 at source S of drive transistor TD through the Data line and through transistor TS2, which is on. This establishes a known gate-source voltage Vgs across drive transistor TD. The known Vgs value and the threshold voltage Vt of transistor TD determine the amount of current flowing through diode 44 from the Data line. Display driver circuitry 20 measures this current during period 208 to ascertain the value of threshold voltage Vt. Appropriate threshold voltage compensation operations may then be performed by adjusting the values of the image data loaded into pixel 22 during data writing operations 202 (FIG. 22).

(90) If desired, a settling time may be inserted into the process of FIG. 20A as illustrated in FIG. 20B. The settling time allows the voltage on data line Data to be established at a high voltage near to Vddel to allow light-emitting diode 44 to mimic normal emission operations during current sensing. Sensing settling operations allow analog-to-digital converter circuitry in circuitry 20 that is coupled to data line Data sufficient time to sample the voltage on line Data.

(91) FIG. 23 shows an illustrative 6T1C configuration for pixel 22. Transistor TS3 and transistor TS2 may be controlled by scan signal Scan2 as shown in FIG. 23, or the gate of transistor TS3 may be controlled using a previous scan line signal (e.g., Scan2(n-1) from a previous row). Transistor TS3 in FIG. 23 may be used to reset Node4 at the anode of light-emitting diode 44. The parasitic capacitance of light-emitting diode 44 can discharge Node4 rapidly (e.g., from about 2.5 volts to -6 volts) to turn off light-emitting diode 44 quickly during data writing. This helps lower Node4 below the threshold voltage of light-emitting diode 44 and helps prevent light-emitting diode 44 from turning on due to leakage from drive transistor TD during the displaying of black images on display 14. FIG. 24 show illustrative control signals that may be used in operating pixel 22 of FIG. 23 during on-bias stress, data writing, and emission periods.

(92) In the illustrative configuration for pixel 22 of FIG. 25, TS3 has been replaced by bypass transistor TS4 (controlled by Scan3) to help prevent current from passing through transistor TD and undesirably illuminating diode 44 while performing current sensing operations on transistor TD. If desired, transistor TS4 may be placed in alternate location TS4'. The example of FIG. 25 is merely illustrative. FIG. 26 shows control signals that may be used in operating pixel 22 of FIG. 25. FIG. 27 shows pixel 22 of FIG. 25 during on-bias stress operations. FIG. 28 shows pixel 22 of FIG. 25 during data writing. FIG. 29 shows pixel 22 of FIG. 25 during emission operations. FIG. 30 shows pixel 22 of FIG. 25 during current sensing operations to measure Vt of TD (in which light-emitting diode 44 is not turned on due to the current bypass path established by transistor TS4. In the example of FIG. 31, transistor TS4 is being used in a voltage sensing scheme. In the voltage sensing scheme of FIG. 31, transistor TS3 is used to avoid creating a voltage drop over transistors TS2, TD, and TE2 to enhance sensing accuracy.

(93) FIG. 32 is a diagram of the type shown in FIG. 26 showing how current sensing operations of the type described in connection with FIG. 30 may be performed.

(94) As these examples demonstrate, an additional transistor may be incorporated into pixel 22 to create a current bypass path during threshold voltage measurements on drive transistor TD. Because the additional transistor is used in creating a bypass path that bypasses light-emitting diode 44, the additional transistor may sometimes be referred to as a bypass transistor. The bypass transistor may be, for example, a silicon transistor (i.e., a transistor with a silicon active region).

(95) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A display comprising a plurality of pixels, wherein a pixel in the plurality of pixels comprises: a first transistor; a second transistor; a third transistor; a light-emitting diode, wherein the first transistor, the second transistor, the third transistor, and the light-emitting diode are connected in series between a first power supply terminal and a second power supply terminal; a fourth transistor that is coupled to a source of the second transistor; and a fifth transistor that is coupled to an anode of the light-emitting diode, wherein a gate of the fifth transistor and a gate of the fourth transistor receive a common control signal.
2. The display defined in claim 1, wherein the pixel comprises: a sixth transistor that is coupled to a gate of the second transistor.
3. The display defined in claim 2, wherein the sixth transistor is coupled between the gate of the second transistor and a signal line, and wherein the fifth transistor is coupled between the anode of the light-emitting diode and the signal line.
4. The display defined in claim 1, wherein the pixel comprises: a capacitor coupled to a gate of the second transistor.
5. The display defined in claim 1, wherein the second transistor is a drive transistor and wherein the first and third transistors are emission transistors.
6. The display defined in claim 5, wherein a gate of the first transistor and a gate of the third transistor receive a common emission control signal.
7. The display defined in claim 1, wherein the second transistor is connected between the first and third transistors.
8. The display defined in claim 1, wherein the second transistor is a drive transistor and the fourth and fifth transistors are switching transistors.
9. The display defined in claim 1, wherein the first, second, third, fourth, and fifth transistors are p-type silicon transistors.
10. The display pixel defined in claim 1, wherein the fourth transistor is coupled between the source of the second transistor and a data line and wherein the fifth transistor is coupled between the anode of the light-emitting diode and a signal line.
11. A display pixel comprising: a first power supply terminal; a second power supply terminal; a first transistor, a second transistor, a third transistor, and a light-emitting diode that are connected in series between the first power supply terminal and the second power supply terminal; a fourth transistor that is coupled between a gate of the second transistor and a signal line; a fifth transistor that is coupled between a source of the second transistor and a data line; and a sixth transistor that is coupled between an anode of the light-emitting diode and the signal line.
12. The display pixel defined in claim 11, wherein the first and third transistors are emission transistors and wherein the second transistor is a drive transistor connected between the first and third transistors.
13. The display pixel defined in claim 11, wherein the second transistor is a drive transistor and the fourth, fifth, and sixth transistors are switching transistors.
14. The display pixel defined in claim 11, wherein the first, second, third, fifth, and sixth transistors are silicon transistors.
15. The display pixel defined in claim 11, further comprising: a capacitor that is coupled to the gate of the second transistor.
16. A display pixel comprising: a first power supply terminal; a second power supply terminal; a first transistor, a second transistor, a third transistor, and a light-emitting diode that are connected in series between the first power supply terminal and the second power supply terminal; a fourth transistor that is coupled to a gate of the second transistor; a fifth transistor that is coupled between a source of the second transistor and a data line; and a sixth transistor that is coupled to an anode of

the light-emitting diode, wherein the first, second, third, fifth, and sixth transistors comprise p-channel silicon transistors.

17. The display pixel defined in claim 16, further comprising: a capacitor coupled to the gate of the second transistor.

18. The display pixel defined in claim 16, wherein the fourth transistor is coupled between the gate of the second transistor and a signal line.

19. The display pixel defined in claim 18, wherein the sixth transistor is coupled between the anode of the light-emitting diode and the signal line.

20. The display pixel defined in claim 16, wherein the first and third transistors are emission transistors, and wherein a gate of the first transistor and a gate of the third transistor receive a common emission control signal.
